Dual Spatial–Spectral Pyramid Network With Transformer for Hyperspectral Image Fusion

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***Abstract*— Multispectral image (MSI) and hyperspectral image (HSI) fusion can combine the best of both worlds to produce images with both high spatial and spectral resolutions. In this article, we have designed a network for fusing MSIs and HSIs, called DSPNet. On the one hand, in order to ensure the accuracy of the spectral dimension, i.e., spectral fidelity, we designed the spectral pyramid (SpePy) module and the multiscale local spectral information fusion (MLSIF) module. The former extracts the multiscale local spectral information that captures the subtle spectral details and variations between different spectra. The latter establishes long-range dependency in the spectral dimension through the spectralwise multihead hybrid-attention (S-MHA) mechanism, thus enabling the network to focus on the local spectral information needed to recover the spectral details. On the other hand, to address the spatial information of MSIs, we designed the spatial pyramid (SpaPy) module. The SpaPy module can extract the nonlocal spatial information of MSIs at different scales, which enables the network to adapt to different remote sensing scenes. Experiments performed on simulated and real data demonstrate the superiority of our method over the state-of-the-art methods both qualitatively and quantitatively. Our code is publicly available at https://github.com/syc11-25/DSPNet.**

***Index Terms*— Hyperspectral image (HSI) fusion, local spectral information, multiscale, pyramid structure, spectralwise Transformer.**

I. Introduction

HYPERSPECTRAL imaging is a powerful technique for acquiring detailed spectral information about a target scene. With its ability to capture images in tens or even hundreds of narrow and continuous spectral bands, HSIs are made to obtain critical spectral information about the ground surface. Therefore, HSIs shine in many remote sensing fields, such as classification [1], target detection [2], vegetation monitoring [3], and water quality monitoring [4]. In contrast, multispectral imaging captures images in a few broad spectral bands, providing higher spatial resolution but sacrificing spectral resolution. The tradeoff between spatial and spectral resolution poses a challenge for remote sensing applications, as some applications may require both high spatial and spectral resolutions.

The potential benefits of fusing high-resolution multispectral images (HR-MSIs) with low-resolution hyperspectral images (LR-HSIs) have motivated researchers to develop and explore different methods for image fusion. Existing MSI and HSI fusion methods [5] have demonstrated on simulated and real data that image fusion can effectively improve the spatial resolution of HSIs and demonstrate advantages in areas such as fine classification of surface objects [6], the urban change detection [7], and the soil composition estimation [8].

Reviewing the developments in the past few years, MSI and HSI fusion methods can be broadly classified into three categories: pansharpening-based approaches, decomposition-based approaches, and deep learning-based approaches.

Pansharpening-based approaches draw on the panchromatic sharpening task, which implements multispectral and panchromatic image fusion or hyperspectral and panchromatic image fusion [9], [10]. This task can be seen as a special case of MSI and HSI fusion tasks, where the panchromatic images provide spatial information, while the MSIs or HSIs provide spectral information. The pansharpening-based approaches can be subdivided into component substitution (CS)-based methods [11] and multiresolution analysis (MRA)-based methods [12], [13]. CS-based methods separate the spatial and spectral information of the upsampled HSI into different components and replace the spatial component with the corresponding MSI. MRA-based methods, on the other hand, inject high-resolution information from the MSI, obtained through multiresolution decomposition [14], [15], [16], into LR-HSI.

Decomposition-based approaches model HSIs and MSIs in different ways, mainly using matrix or tensor decomposition-based methods. The former unfolds HSIs and MSIs into 2-D matrices and considers them decomposable into spectral bases and corresponding coefficients. The fusion is then accomplished by sparse decomposition or low-rank methods. However, this approach may lose some information and fail to fully account for spatial–spectral correlation. The latter models the HSIs and MSIs in a tensor fashion, which solves this problem to some extent. Tensor decomposition-based methods consider that HSIs and MSIs can be represented by dictionaries

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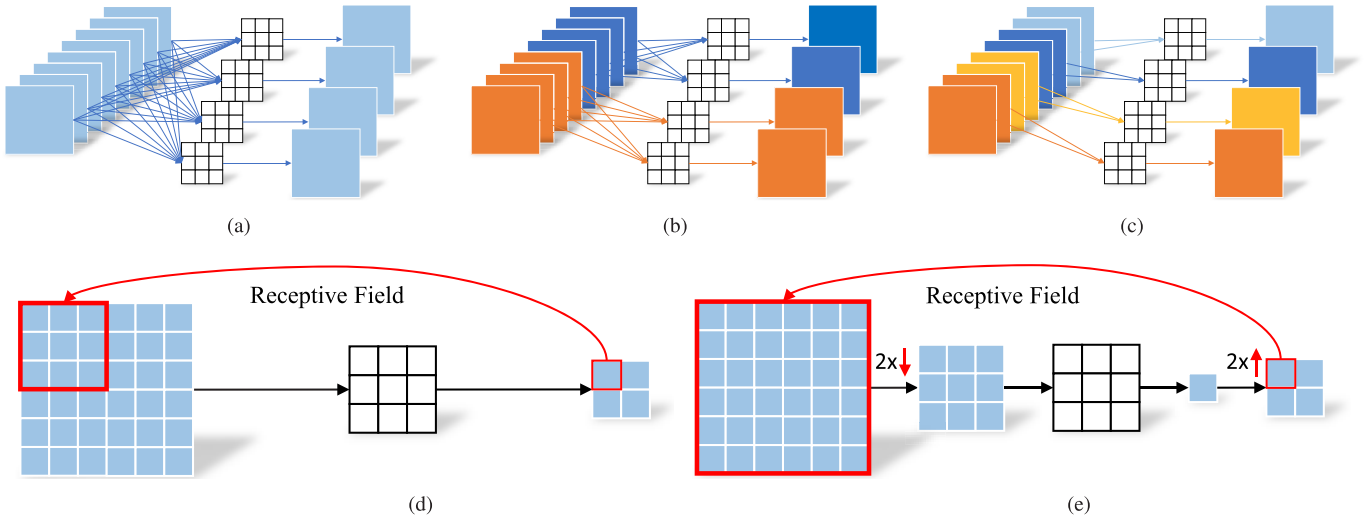


Fig. 1. Illustration of the difference between (a) global spectral information extraction and (b) and (c) multiscale local spectral information extraction ($G = 2$ and $G = 4$), respectively, and the difference between (d) local spatial information extraction and (e) nonlocal spatial information extraction).

of widths, heights, and spectral modes as well as a tensor core, and some learning dictionary algorithms are used to learn these dictionaries.

The deep learning-based approaches have grown rapidly and shown great advantages. These methods utilize various network architectures such as two-branch convolutional neural network (CNN) [17], generative adversarial network (GAN) [18], and Transformer [19], [20] to establish mapping relationships between LR-HSI, HR-MSI, and HR-HSI. The deep learning-based approach obtains spatial features of HR-MSI and spectral features of LR-HSI and fuses them together using appropriate constraints. Common constraints for training networks are loss functions, such as mean absolute error (MAE) loss and spectral angle mapper (SAM) loss. These methods have shown promising fusion results on large datasets and have been widely applied to various fields.

Despite the remarkable achievements of deep learning-based methods, there is still room for improvement in extracting valuable information from HSIs and MSIs. First, the existing methods cannot solve the difficulty of extracting spectral information well due to the large number of spectra. Most methods directly process all the spectra of HSIs together. Fig. 1(a) shows how each convolution kernel in the convolutional layer operates on all spectral channels simultaneously. This makes it difficult to extract sufficiently fine information from HSIs with a high number of spectra. Also, the negative impact of spectra that contain less information on feature extraction is always present. Second, real remote sensing scenes, such as aerial remote sensing images and satellite images, tend to take an overhead view of a larger scene, which often contains objects of various scales. Existing methods lack targeted design for these problems and only extract features with local receptive fields, making it difficult to effectively extract global scene features and features of objects at various scales. Although the existing methods [21], [22] try to inject the spatial information in the encoding stage or decoding stage, the local convolution they employ limits their effectiveness. The local convolution has a fixed and small receptive field, as shown in Fig. 1(d), which is difficult to cope with scenes with large ranges or scenes containing objects of different sizes.

To tackle the challenges mentioned earlier, we propose a novel three-branch dual pyramid network with spectralwise Transformer for MSI and HSI fusion, namely, DSPNet. First, we use the idea of group convolution to process HSIs to extract local spectral information. In order to more accurately grasp the variation of spectral dimension, we also introduce the idea of multiscale based on the band grouping and design the spectral pyramid (SpePy) module to extract the multiscale local spectral information. As shown in Fig. 1(b) and (c), we use different convolution kernels to extract information from the grouped bands. This not only extracts more detailed spectral features but also filters out the influence of other bands. Second, in order to fuse the multiscale local spectral information with the global spatial–spectral information extracted by spatial–spectral branch, we propose a multiscale local spectral information fusion (MLSIF) module. We leverage the spectralwise Transformer to establish long-range dependency between the spectral information at each scale. Guided by the spectralwise multihead hybrid-attention (S-MHA), the network not only recorrelates the grouped spectra to ensure the continuity of spectral dimensions but also dynamically establishes the weight assignment between spectral groups to ensure the full utilization of significant spectral information and suppression of redundant spectral information. Finally, we design the spatial pyramid (SpaPy) module to extract multiscale nonlocal spatial information. Scenes in remote sensing images are usually large in size and often contain similar but not adjacent detail information, making remote sensing images nonlocal. Therefore, we obtain this nonlocal information by expanding the receptive field to effectively cope with complex remote sensing environments with large scenes and numerous objects. Fig. 1(d) and (e) shows that the extraction of nonlocal spatial information is effective in enhancing the receptive field and enabling the network to focus on a larger scene. The introduction of a multiscale design can enable us to capture spatial characteristics using receptive

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fields of different sizes simultaneously. The contributions of our method are summarized in the following three aspects.

1) We propose the SpePy module to enhance the spectral fidelity. Benefiting from the multiscale design and band grouping of spectral dimension, we can characterize the variation of spectral dimensions in a finer and more comprehensive way while also alleviating the difficulties arising from high spectral numbers for information extraction.
2) The proposed MLSIF module effectively fuses multiscale local spectral information by utilizing the S-MHA mechanism. This mechanism can capture long-range dependencies between different spectral groups to recorrelate the grouped spectra and establish weight assignments between them in a displacive manner, resulting in improved fusion performance.
3) We propose the SpaPy module, which emphasizes nonlocal information in the spatial dimension and incorporates a multiscale design. The module addresses complex remote sensing scenarios by utilizing receptive fields of different sizes, enabling the network to capture overall scene information while also focusing on targets of various scales.

The rest of this article is arranged as follows. Section II provides some discussions of work related to MSI and HSI fusion. In Section III, we give a detailed introduction to our proposed DSPNet as well as the loss function used to train the network. In Section IV, qualitative and quantitative comparisons and ablation study are performed. Section V gives the conclusion.

II. Related Work

In this section, we give a brief introduction to the pansharpening-based methods, decomposition-based methods, and deep learning-based fusion methods.

A. Pansharpening-Based Methods

The task of pansharpening is a special example of MSI and HSI fusion, so the extension of the method used for pansharpening to MSI and HSI fusion became an early research focus. Considering that panchromatic images are single-band images, while both MSI and HSI are multiband, methods used for pansharpening are generally extended to MSI and HSI fusion by means of band selection. These papers [23], [24] use the similarity measure of the relative spectral response of MSI and HSI sensors to divide the HSI into multiple band groups in the spectral dimension and correspond each band of the MSI to each band group of the HSI. The method [25], on the other hand, performs the similarity measure directly on images and then performs the band selection so that different image pairs from the same pair of sensors also produce different ways of band assignment. Also, all HSI bands in each group are enhanced using the same MSI band. Specifically, fusion is achieved for each group and its corresponding MSI band using either a CS-based or MRA-based approach. For example, Vivone [26] improves on the classical CS-based approach, BDSD, to accommodate the case where MSI has more than four bands. Vivone et al. [27] used an MRA-based approach to extract high-frequency spatial detail information and estimate the injection coefficients on the full scale.

B. Decomposition-Based Methods

The decomposition-based approach formulates the fusion problem as an optimization problem and has been widely studied in recent years. The matrix-based decomposition approach transforms the problem into a problem of estimating the spectral basis D and coefficients A in 2-D space. The method [28] treats the spectral basis D as an overcomplete dictionary and learns this dictionary from HSIs by a sparse dictionary learning algorithm. Other methods [29], [30] use methods such as vertex component analysis or truncated singular value decomposition to learn spectral basis D from HSIs. Since MSI and HSI can be considered as a 3-D cube, they can be fused using a method based on tensor decomposition. Dian et al. [31] treated HSI as a tensor and decomposed it into a kernel tensor and the dictionaries of three dimensions using Tucker decomposition and then processed these components with the help of a nonlocal approach. Similarly, in the literature [32], the method of coupled sparse tensor decomposition is used to decompose the LR-HSI and HR-MSI and redefine the fusion problem as the estimation of a core tensor and dictionaries of the three modes. Compared with the matrix decomposition-based method, the tensor decomposition-based method can more fully consider the correlation between the spatial and spectral information of the source images. However, such methods require accurate estimation of the point spread function and the spectral response function of the sensor, which limits the generalizability of such methods.

C. Deep Learning-Based Methods

In recent years, deep learning-based methods have evolved rapidly and have shown great advantages over traditional methods in numerous low-level vision tasks [33]. Xiao et al. [21] used dual UNet to extract the depth features of both MSI and HSI and then gradually injected the spatial information of MSI into HSI. The network gradually fused the spatial and spectral information through three steps of directly fusion, slow fusion, and late fusion, and finally obtained HR-HSI. SGARDN [34] explores the application of group convolution in single HSI super-resolution tasks as a way to avoid the spectral disorder caused by conventional convolution. With the advantage of group convolution, SGARDN achieves super-resolution reconstruction of HSIs with high accuracy. Both DualSR [35] and PriNET+ [36] better enable HSI reconstruction tasks by exploring the spectral correlation between neighboring bands in a large spectral range. The outstanding results of these works demonstrate the importance of exploring the correlation between spectra. PSRT [37] proposes a new Transformer-based network framework that combines shuffle-and-reshuffle (SaR) strategies and multiscale feature extraction to learn local and more distant representations. The customized SaR strategy improves the efficiency of modeling long-distance dependency by propagating information between different windows and greatly

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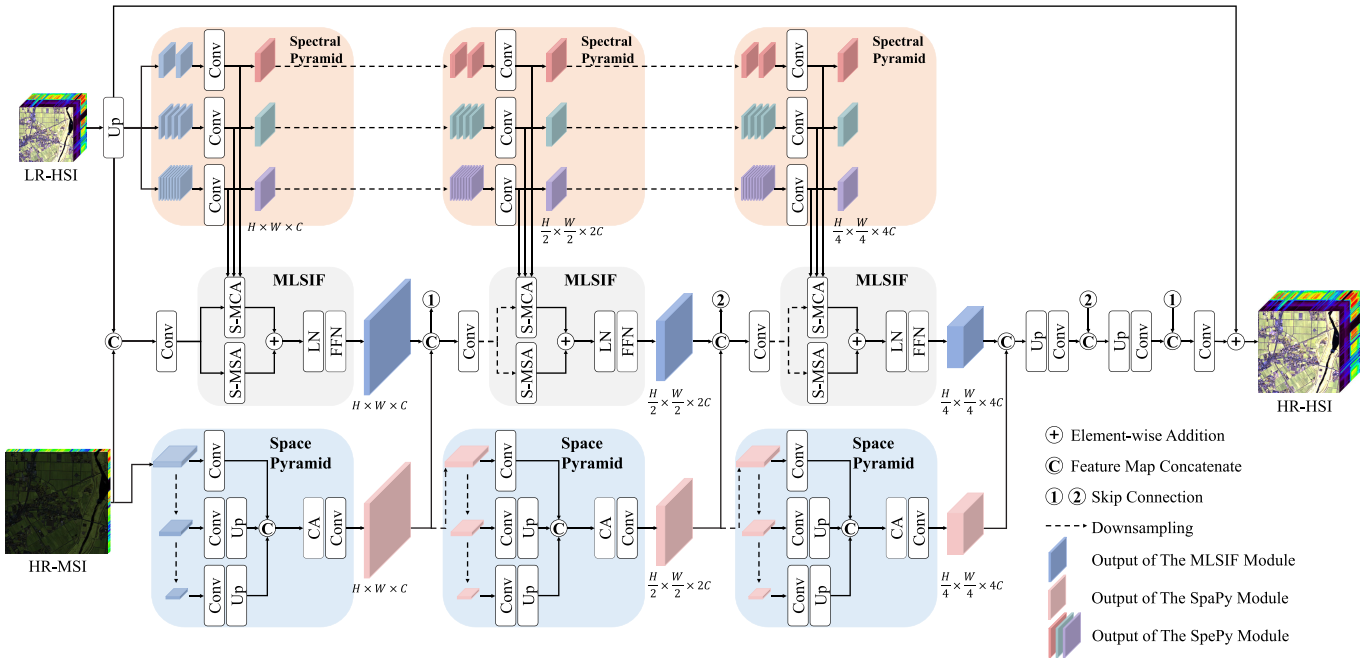


Fig. 2. Overall framework of the proposed DSPNet. The framework from top to bottom is the spectral branch, the spatial–spectral branch, and the spatial branch. The input LR-HSI is upsampled to the same size as the HR-MSI by means of bicubic interpolation. H, W, and C denote the height, width, and number of channels of the feature maps, respectively.

reduces the amount of computation. MCT-NET [38] proposes a multihierarchical Transformer, which adds the idea of cross-attention to the traditional Transformer's self-attention mechanism to better achieve spatial–spectral information retention of fused images. Multiscale spatial–spectral Transformer network (MSST-Net) [39] integrates the self-attention mechanism of Transfomer and extracts the spectral features of HSI and the spatial features of MSI in two branches. With the self-attention mechanism and multiscale design, MSST-Net achieves good results in both full-resolution experiments and reduced-resolution experiments. MoG-DCN [40] is model-guided image super-resolution algorithm that considers both domain knowledge likelihood and deep image prior. By fully integrating the spectral observation model with neural networks, MoG-DCN achieves good results. Huang et al. [41] proposed the DHIF-Net model, which is based on the spatially adaptive 3-D filter to fully exploit the spatiospectral dependency. Based on the proposed spatial–spectral regularization, the HSI fusion objective function is differentiable, and thus, end-to-end optimization of the iterative spatial–spectral regularization can be performed through a multilevel network. Despite the excellent performance achieved by existing methods, they lack targeted design for spectral information and complex remote sensing scenarios. Motivated by this, we propose DSPNet by introducing multiscale design of spectral and spatial dimensions and hybrid-attention mechanism in the network.

III. Proposed Method

In this section, the problem formulation is provided, followed by a detailed description of our proposed network framework as well as the loss function.

A. Problem Formulation

As shown in Fig. 2, the DSPNet extracts the spatial and spectral information of the source images using a three-branch network, where LR-HSI $\in \mathbb{R}^{h \times w \times B}$ (h, w, and B represent the height, width, and band number, respectively), HR-MSI $\in \mathbb{R}^{H \times W \times b}$, and HR-HSI $\in \mathbb{R}^{H \times W \times B}$. We denote the space size ratio of HR-MSI and LR-HSI by r. Then, $H = h \times r$ and $W = w \times r$. First, we input the upsampled LR-HSI into the SpePy module to extract the multiscale local spectral information. The upsampled LR-HSI and HR-MSI are concatenated and fed into a convolutional layer with a kernel size of 3 and a stride of 1 to obtain the spatial–spectral information. The HR-MSI is fed into the SpaPy module to extract nonlocal spatial information (i.e., pink blocks). Then, the multiscale local spectral information and spatial–spectral information are fused in the MLSIF module. The fused feature maps (i.e., blue blocks) are concatenated with the output of the SpaPy module and fused by convolution with a downsampling operation. The fusion result is fed as input to the next MLSIF module. Finally, we reconstruct the high-resolution spatial and spectral information using continuous convolution with upsampling operations. During the reconstruction process, in order to aggregate the contextual information, we introduce the skip connection operation. The output of the spatial–spectral branch is added with the upsampled HSI by a skip connection, and the summed results are the HR-HSI. This operation allows the network to learn only the residual information at the target resolution so that a concise encoder architecture is also sufficient for the reconstruction task.

B. SpePy Module

The SpePy module is used to extract the multiscale local spectral information of HSI, and its architecture is shown in

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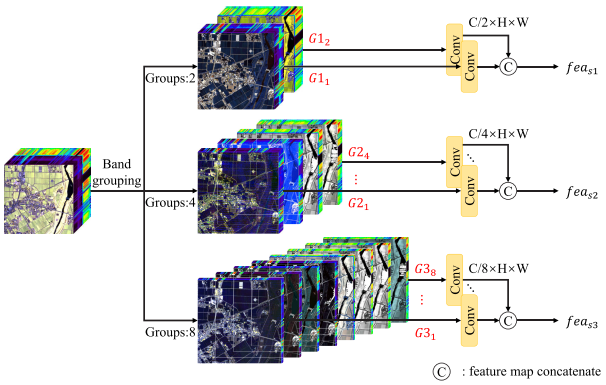


Fig. 3. Architecture of the SpePy module. The input is the HSI or feature maps. C, H, and W denote the number of channels, height, and width of the feature maps output, respectively. Gi_j denotes the jth spectral group of the ith group. $\text{fea}_{\text{si}} \in \mathbb{R}^{C\times H\times W}$ denotes the feature maps of the local spectra at different scales, $i \in \{1, 2, 3\}$.

Fig. 3. First, the HSI is divided into three groups, namely, $G1$–$G3$, by different band grouping methods. Then, we extract the features of these spectral groups separately by using the group convolution with group numbers of 2, 4, and 8. In group convolution, different spectral groups Gi_j in Gi are processed by different convolution kernels, and the outputs of these convolution kernels are concatenated within the group before being output. The formulas for group convolution are shown as follows:

$$\begin{aligned} \text{fea}_{\text{si}_j} &= \sigma\left(Wi_j * Gi_j\right), \quad 1 \le j \le J \\ \text{fea}_{\text{si}} &= \text{cat}\left(\text{fea}_{\text{si}_1}, \ldots, \text{fea}_{\text{si}_J}\right), \quad J = 2^i \end{aligned} \tag{1}$$

where $*$ is the convolution operator, Wi_j are the convolution kernels that processes Gi_j, σ is the activation function, and cat is the concatenation operator. By the way, the input of the second and third SpePy module is no longer the HSI, but the output fea_{s1}, fea_{s2}, and fea_{s3} of the previous SpePy module treat them as $G1$–$G3$, respectively. The rest of the processing is the same. If the number of bands in the input image or feature maps is not a multiple of 8, the missing bands are added by duplicating the last few bands. The SpePy module can fully exploit the local spectral information at different scales. This multiscale local spectral information is sent to the MLSIF module as input. Also, the reasonable use of this local spectral information can help the network learn and reconstruct more detailed spectral information.

C. Multiscale Local Spectral Information Fusion Module

This section introduces how we can take advantage of the extracted multiscale local spectral information. The SpePy module artificially divides HSIs by band grouping, and these groups have information that varies from each other, guiding the image to improve resolution and spectral fidelity. To make this multiscale local spectral information useful to a great extent, we design the MLSIF module to guide the spatial–spectral branching to fuse this information. The architecture is shown in Fig. 4. The design of the module is based on the structure of the Transformer. The difference is that we replace the multihead self-attention (MSA) mechanism in the classical Transformer with a S-MHA mechanism. S-MHA treats each spectral feature map as a token and computes spectralwise multihead self-attention (S-MSA) and spectralwise multiheaded cross-attention (S-MCA) between spectral features at different scales along the spectral dimension. First, we reshape the inputs fea_g and $\text{fea}_{\text{si}} \in \mathbb{R}^{H\times W\times C}$ into spectral tokens $\hat{\text{fea}}_g$ and $\hat{\text{fea}}_{\text{si}} \in \mathbb{R}^{HW\times C}$, respectively. Then, using the learnable weights $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V \in \mathbb{R}^{C\times C}$, we linearly project them into spectral query $\mathbf{Q}$, spectral key $\mathbf{K}$, and spectral value $\mathbf{V} \in \mathbb{R}^{HW\times C}$. The formulas are given as follows:

$$\begin{aligned} \left\{\mathbf{Q}_g, \mathbf{K}_g, \mathbf{V}_g\right\} &= \left\{\hat{\text{fea}}_g\mathbf{W}_g^Q, \hat{\text{fea}}_g\mathbf{W}_g^K, \hat{\text{fea}}_g\mathbf{W}_g^V\right\} \\ \left\{\mathbf{K}_{\text{si}}, \mathbf{V}_{\text{si}}\right\} &= \left\{\hat{\text{fea}}_{\text{si}}\mathbf{W}_{\text{si}}^K, \hat{\text{fea}}_{\text{si}}\mathbf{W}_{\text{si}}^V\right\}. \end{aligned} \tag{2}$$

After that, the hybrid-attention function computes the dot product of the queries of $\hat{\text{fea}}_g$ and keys of $\hat{\text{fea}}_g$ and $\hat{\text{fea}}_{\text{si}}$, which is then normalized using the softmax function to obtain the self-attention map A_g and cross-attention map A_{si}

$$\begin{aligned} A_g &= \text{softmax}\left(\sigma_g\mathbf{K}_g^T\mathbf{Q}_g\right), \quad \text{head}_g = \mathbf{V}_g A_g \\ A_{\text{si}} &= \text{softmax}\left(\sigma_{\text{si}}\mathbf{K}_{\text{si}}^T\mathbf{Q}_g\right), \quad \text{head}_{\text{si}} = \mathbf{V}_{\text{si}} A_{\text{si}} \end{aligned} \tag{3}$$

where $\mathbf{K}_g^T$ and $\mathbf{K}_{\text{si}}^T$ are the transposed matrix of $\mathbf{K}_g$ and $\mathbf{K}_{\text{si}}$, respectively, and σ_g and $\sigma_{\text{si}} \in \mathbb{R}^1$ are the learnable parameters to characterize the effect of wavelength in the spectral dimension. head_g and head_{si} are the corresponding self-attention scores and cross-attention scores, respectively. In order for the attention mechanism to capture information from different representation subspaces simultaneously, we extend the hybrid attention into multihead hybrid attention by referring to the literature [42]. Thus, for h headers, we get the h-group output, which we treat as follows:

$$\begin{aligned} \text{S-MSA}\left(\mathbf{Q}_g, \mathbf{K}_g, \mathbf{V}_g\right) &= \text{cat}\left(\text{head}_g^1, \ldots, \text{head}_g^h\right)\mathbf{W} + \text{pos}\left(\mathbf{V}_g\right) \\ \text{S-MCA}_i\left(\mathbf{Q}_g, \mathbf{K}_{\text{si}}, \mathbf{V}_{\text{si}}\right) &= \text{cat}\left(\text{head}_{\text{si}}^1, \ldots, \text{head}_{\text{si}}^h\right)\mathbf{W} \\ \text{S-MHA} &= \text{S-MSA} + \sum_{i=1}^{3}\text{S-MCA}_i \end{aligned} \tag{4}$$

where $\mathbf{W} \in \mathbb{R}^{C\times C}$ are the learnable weights and $\text{pos}(\cdot)$ that consists of two convolutional layers with gaussian error linear unit (GELU) activation layer is deployed to generate the position embedding, which is used to provide position information that is missing from the attention mechanism. Next, in order to refine the features yielded by the attention mechanism, the output of S-MHA is sent to a layer normalization (LN) and a feedforward network (FFN). The LN is used to improve the structure of the network. We build this FFN with two 1×1 and one 3×3 convolutional layers with GELU activation layers. Thus, the overall flow of the MLSIF module can be expressed as follows:

$$\begin{aligned} \left\{\mathbf{Q}_g, \mathbf{K}_g, \mathbf{V}_g\right\} &= \left\{\hat{\text{fea}}_g\mathbf{W}_g^Q, \hat{\text{fea}}_g\mathbf{W}_g^K, \hat{\text{fea}}_g\mathbf{W}_g^V\right\} \\ \left\{\mathbf{K}_{\text{si}}, \mathbf{V}_{\text{si}}\right\} &= \left\{\hat{\text{fea}}_{\text{si}}\mathbf{W}_{\text{si}}^K, \hat{\text{fea}}_{\text{si}}\mathbf{W}_{\text{si}}^V\right\}, \\ \text{S-MSA}\left(\mathbf{Q}_g, \mathbf{K}_g, \mathbf{V}_g\right) &= \text{cat}\left(\text{head}_g^1, \ldots, \text{head}_g^h\right)\mathbf{W} + \text{pos}\left(\mathbf{V}_g\right) \\ \text{S-MCA}_i\left(\mathbf{Q}_g, \mathbf{K}_{\text{si}}, \mathbf{V}_{\text{si}}\right) &= \text{cat}\left(\text{head}_{\text{si}}^1, \ldots, \text{head}_{\text{si}}^h\right)\mathbf{W} \\ \text{S-MHA} &= \text{S-MSA} + \text{S-MCA}_1 \\ &\quad + \text{S-MCA}_2 + \text{S-MCA}_3 \\ O &= \text{reshape}(\text{FFN}(\text{LN}(\text{S-MHA})) \\ &\quad + \text{S-MHA}) \end{aligned} \tag{5}$$

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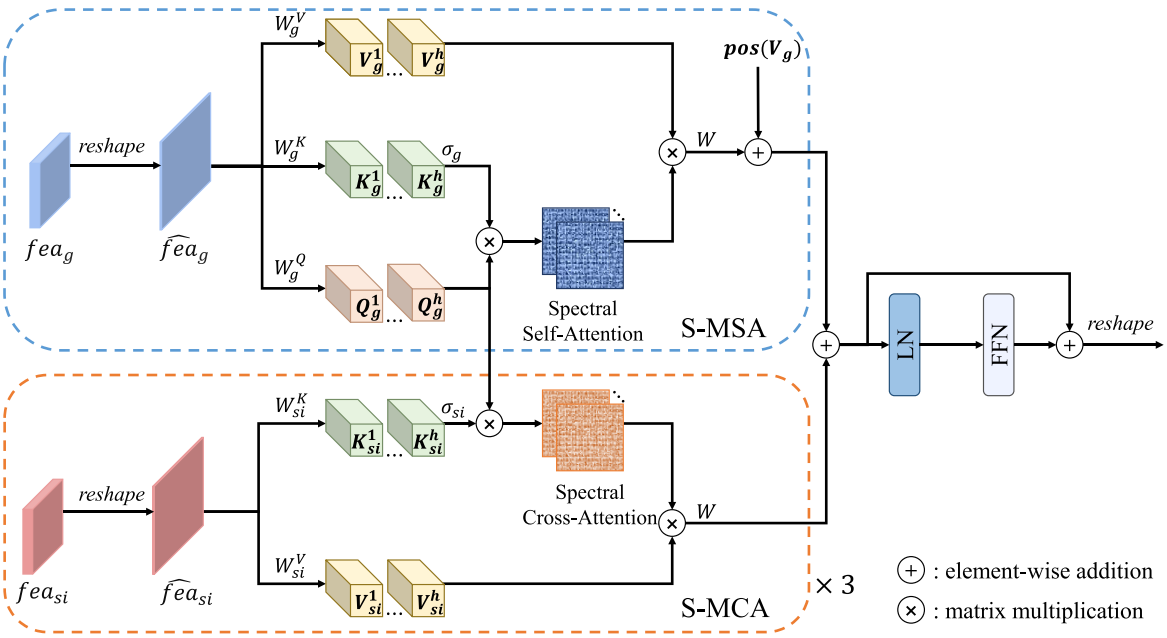


Fig. 4. Architecture of MLSIF module. fea_g is the feature extracted by the convolutional layer in the spatial–spectral branch. $\text{fea}_{si} \in \mathbb{R}^{C \times H \times W}$ denotes the feature maps of the local spectra at different scales, $i \in \{1, 2, 3\}$.

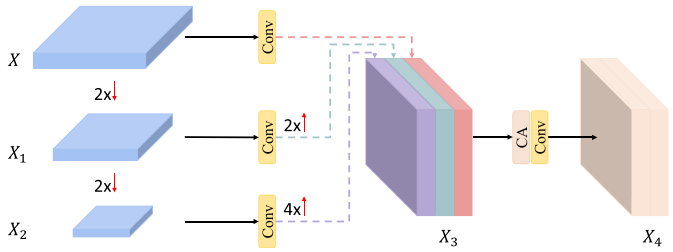


Fig. 5. Architecture of SpaPy module: $2\times\downarrow$ denotes downsampling to half of the original scale, and $2\times\uparrow$ and $4\times\uparrow$ denote upsampling to two and four times of the original scale, respectively. The feature maps pointed by the dashed arrows are concatenated along the channel dimension.

where $O \in \mathbb{R}^{H \times W \times C}$ is the final output of the MLSIF module.

D. Spatial Pyramid Module

The SpaPy module is designed to extract the nonlocal spatial details of the MSI. Generally, the ground objects in remote sensing images vary greatly in scale, and the ground scenes are complex and variable. In consequence, it is significant to pay attention to nonlocal details along with local details. The architecture is shown in Fig. 5. Suppose that the input of the SpaPy module is $X \in \mathbb{R}^{H \times W \times C}$. We first downsample X twice consecutively to obtain $X_1 \in \mathbb{R}^{(H/2)\times(W/2)\times C}$ and $X_2 \in \mathbb{R}^{(H/4)\times(W/4)\times C}$. Then, we use three convolutional layers with the same convolutional kernel size to extract the spatial information of the input at each scale. Despite having the same size of the convolutional kernels, the perceptual fields of these three convolutional layers are not the same but increase sequentially due to the downsampling operation of the inputs, that is, the first convolutional layer outputs local spatial information and the remaining two convolutional layers output nonlocal spatial information. The output feature maps are upsampled to the original scales and then concatenated together to obtain $X_3 \in \mathbb{R}^{H \times W \times 3C}$. Later, we use the channel attention (CA) mechanism [43] to refine the whole feature map. The CA mechanism adjusts the weights of each channel of the feature maps in the channel dimension, thus facilitating the representation of useful information at each scale and the subsequent adaptive fusion. Finally, we use a convolutional layer to integrate these local and nonlocal spatial details and obtain the output of the SpaPy module $X_4 \in \mathbb{R}^{H \times W \times 2C}$. The output of the SpaPy module is not only sent to the spatial–spectral branch for information integration but also downsampled and sent as input to the next SpaPy module.

E. Loss Function

Our network learns the residuals between the ground truth and the input upsampled HSI, i.e., the high-frequency information that the input upsampled HSI lacks. Compared with L_1 loss, L_2 loss is not robust enough and is more sensitive to anomalous samples, so it is not suitable for learning high-frequency information. Moreover, the L_1 loss converges faster, so we choose the L_1 loss between the fused image F and ground truth X as the constraint of the network. The formula is given as follows:

$$L = \frac{1}{\text{HWB}} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{k=1}^{B} \| F_{i,j,k} - X_{i,j,k} \|_1 \tag{6}$$

where i, j, and k represent the row, column, and band wherein the pixel is located, respectively.

IV. Experimental Results

In this section, we first describe the experimental settings, including the datasets, data simulation, and implementation details. Then, we introduce some algorithms used to compare with our proposed DSPNet and the evaluation metrics. Then, qualitative results and quantitative results on four popular used datasets are shown. Finally, ablation experiments are performed to validate the effectiveness of our proposed modules.

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TABLE I
SUMMARY OF THE DATASETS IN TERMS OF THE NUMBER, SIZE, AND WAVELENGTH RANGE

Dataset	Number	Size	Range (nm)
CAVE [44]	32	512 × 512 × 31	400-700
Harvard [45]	50	1040 × 1392 × 31	400-700
Chikusei [46]	1	2517 × 2335 × 128	363-1018
Pavia Centre	1	1096 × 1096 × 102	-

A. Experimental Settings

1) Datasets: The comparison experiments are conducted on four popular datasets, including CAVE,[1] Harvard,[2] Chikusei,[3] and Pavia Centre.[4] The CAVE dataset provided in [44] contains 32 pairs of HSIs with a size of 512 × 512 × 31 and corresponding MSIs, which are used to display a wide variety of real-world materials and objects. We select the first 20 HSIs as the training dataset, and the remaining 12 HSIs constitute the test dataset. The Harvard dataset [45] contains 50 indoor and outdoor HSIs under daylight and 27 HSIs under artificial or mixed illumination. These HSIs cover the spectral range from 420 to 720 nm and have a size of 1392 × 1040 × 31. Similarly, we select the first 30 HSIs under daylight as the training dataset and the remaining 20 HSIs as the test dataset. In addition, we evaluate the performance in remote sensing HSI. The Chikusei dataset [46] captures the agricultural and urban areas of Chikusei, Ibaraki, Japan. The Chikusei image containing 2517 × 2335 pixels has 128 bands with a spectral range from 363 to 1018 nm. We crop the top-left area of size 1500 × 2000 and divide it into 12 image patches with a size of 500 × 500 to form the training dataset. Eight nonoverlapping image patches of size 500 × 500 constitute the test dataset, which are cropped from the remaining area. The Pavia Centre is a scene acquired by the reflective optics system imaging spectrometer (ROSIS) sensor during a flight campaign over Pavia, Northern Italy. It is a 1096 × 1096 pixels image with 102 spectral bands. We crop nine blocks from it as the training dataset and six blocks as the test dataset, all of size 192 × 192. A brief summary of the four datasets is shown in Table I.

2) Data Simulation: Our proposed method requires a supervised training approach and therefore requires HR-HSI as the ground truth. However, in the real world, HR-HSI is usually unavailable. Therefore, we treat the original HSI as the ground truth and then obtain LR-HSI and HR-MSI by means of data simulation. To obtain the corresponding LR-HSIs, we first blur the HR-HSIs with an $s \times s$ ($s = 4$, 8, and 16) Gaussian filter with a standard deviation of 2 and then downsample the blurred HSIs in both the horizontal and vertical directions with a scaling factor of s. Moreover, HR-MSIs are obtained from HR-HSIs by spectral degradation. For the CAVE and Harvard datasets, we apply the spectral response function of the Nikon D700 camera to generate HR-MSIs, while for the Chikuseis dataset, we use the spectral response function of the Canon EOS 5D Mark II camera to generate HR-MSIs. For the Pavia Centre dataset, we treat the original HSI and MSI as HR-HSIs and HR-MSIs, respectively. We show in Fig. 6 some simulated RGB images from the CAVE, Harvard, Chikusei, and Pavia Centre datasets. In the training process, we randomly cut image patches of size 96 × 96 from HR-HSIs and then obtain LR-HSIs of size $(96/s) \times (96/s)$ and HR-MSIs of size 96 × 96 in the above manner. Data enhancement methods, such as geometric flipping and rotation, are applied in the training process to generate more training samples.

3) Implementation Details: We initialize our network parameters by the method described in [47], and the batch size is set to 24. We used the ADAM optimizer to train our model by setting the learning rate to 3×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. All the experiments are performed on an NVIDIA GTX 1080Ti GPU and 2.40-GHz Intel[5] Xeon[5] E5-2673 v3 CPU.

B. Benchmarking

To show the superiority of our algorithm, we compare it with other state-of-the-art methods. The comparison algorithms include traditional algorithms such as the fast fusion of multiband images based on solving Sylvester equation (FUSE) [48]; the generalized Laplacian pyramid hypersharpening (GLP-HS) method [49]; the coupled sparse tensor factorization (CSTF) method [32]; and deep learning-based methods such as ResTFNet [50], SSRNet [51], D-UNet [21], MoG-DCN [40], DHIF-Net [41], and MSST-Net [39]. To ensure the fairness of the experiments, all deep learning-based methods are trained and tested using the same inputs. Moreover, the hyperparameter settings of the comparison algorithms are kept consistent with the original paper.

C. Quality Metrics

In order to objectively assess the fusion results of different methods, four popular statistics are chosen as quality indexes, including the peak signal-to-noise ratio (PSNR), the SAM, the erreur relative globale adimensionnelle de synthèse (ERGAS), and the structural similarity index measure (SSIM).

1) Peak Signal-to-Noise Ratio: The PSNR reflects the spatial distortion in the fusion process and is defined as follows:

$$\mathrm{PSNR}(X_i, F_i) = 20 \cdot \log_{10} \frac{\max(X_i)}{\sqrt{\mathrm{MSE}(X_i, F_i)}} \tag{7}$$

where X_i and F_i represent the ith band of HR-HSIs and fusion results, respectively, and $\mathrm{MSE}(\cdot)$ is the mean square error function. We calculate the PSNR of each band separately and then take the average value as the final result

$$\mathrm{PSNR}(X, F) = \frac{1}{B} \sum_{i=1}^{B} \mathrm{PSNR}(X_i, F_i) \tag{8}$$

[1]Available at: https://www.cs.columbia.edu/CAVE/databases/multispectral/
[2]Available at: http://vision.seas.harvard.edu/hyperspec/
[3]Available at: https://naotoyokoya.com/Download.html
[4]Available at: https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University

[5]Registered trademark.

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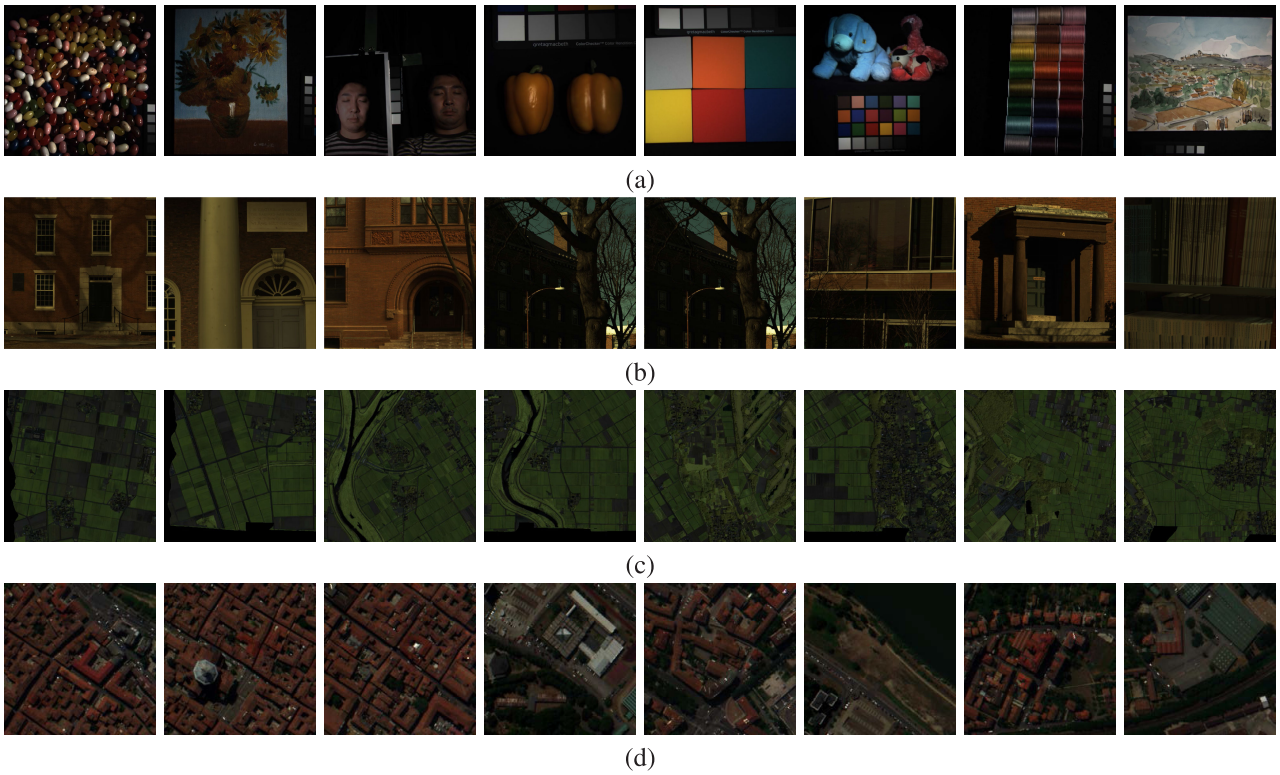


Fig. 6. HR-MSI images from (a) CAVE, (b) Harvard, (c) Chikusei, and (d) Pavia Centre datasets.

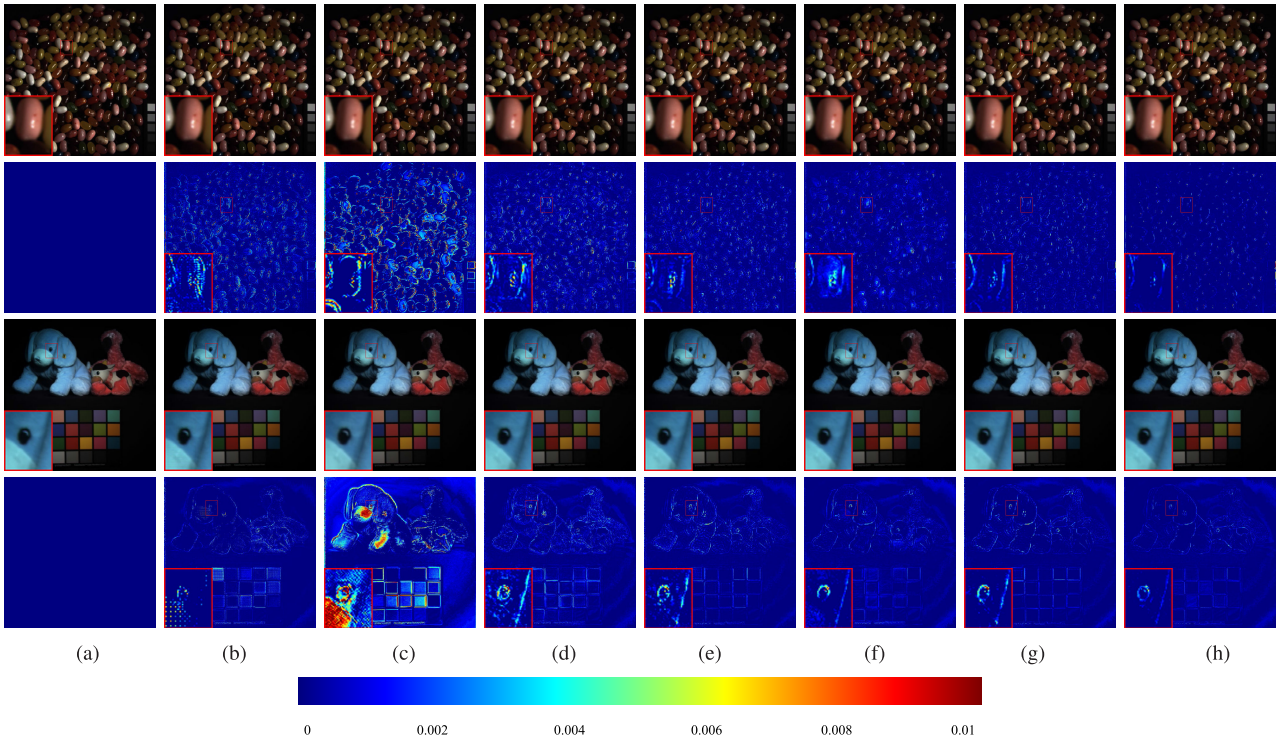


Fig. 7. Qualitative comparison of different methods on the CAVE dataset. The first and third rows show the results using the true-color representation on "jelly_beans" and "stuffed_toys," respectively. The second and fourth rows show the residuals between the GT and the fused results. (a) GT. (b) ResTFNet. (c) SSRNet. (d) D-UNet. (e) MoG-DCN. (f) DHIF-Net. (g) MSST-Net. (h) Ours.

where X and F represent HR-HSIs and fusion results, respectively, and B represents the number of bands. The higher the PSNR, the better the performance.

2) Spectral Angle Mapper: The SAM measures the spectral distortion between HR-HSIs and fusion results. We can calculate it using the following equation:

$$\mathrm{SAM}(X, F) = \frac{1}{\mathrm{HW}} \sum_{i=1}^{\mathrm{HW}} \cos^{-1}\left(\frac{x_i^T f_i}{\|x_i\|_2 \|f_i\|_2}\right) \tag{9}$$

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where $\cos^{-1}$ is the arccosine function; $x_i \in R^{B\times 1}$ and $f_i \in R^{B\times 1}$ represent the spectra of the ith pixel of X and F, respectively; $||\cdot||_2$ is the L_2 norm; and T is the transpose symbol. The smaller SAM indicates better results.

3) Erreur Relative Globale Adimensionnelle De Synthèse : The ERGAS measures the global statistical quality of the fusion result by the following equation:

$$\text{ERGAS}(X, F) = \frac{100}{s}\sqrt{\frac{1}{B}\sum_{i=1}^{B}\frac{\text{MSE}(X_i, F_i)}{\mu_{X_i}^2}} \tag{10}$$

where s refers to the spatial downsampling ratio from HR-HSI to LR-HSI and μ_{X_i} is the mean of X_i. The smaller the ERGAS, the better the fusion result performance.

4) Structural Similarity Index Measure: The SSIM measures the similarity of two images in terms of correlation, brightness, and contrast distortion. The SSIM is defined as follows:

$$\text{SSIM}(X, F) = \frac{1}{B}\sum_{i=1}^{B}\frac{\left(2\mu_{X_i}\mu_{F_i} + C_1\right)\left(2\sigma_{X_iF_i} + C_2\right)}{\left(\mu_{X_i}^2 + \mu_{F_i}^2 + C_1\right)\left(\sigma_{X_i}^2 + \sigma_{F_i}^2 + C_2\right)} \tag{11}$$

where μ_{X_i} and μ_{F_i} are the mean of X_i and F_i, respectively; and σ_{X_i} and σ_{F_i} are the variances of X_i and F_i, respectively. The variable $\sigma_{X_iF_i}$ is the covariance of X_i and F_i. The constants C_1 and C_2 are used to maintain stability and set to 1×10^{-4} and 9×10^{-4}, respectively. The closer the SSIM is to 1, the better the performance.

5) Root-Mean-Square Error: The RMSE is fully known as root-mean-square error and is used to measure the degree of pixel-level similarity between two images. The SSIM is defined as follows:

$$\text{RMSE}(X, F) = \sqrt{\frac{1}{\text{BHW}}\sum_{i=1}^{\text{BHW}}(x_i - f_i)^2} \tag{12}$$

where x_i and f_i represent pixels in images X and F, respectively. The RMSE takes the square root of the error value and hence is less affected by outliers compared to the MSE. The smaller the RMSE, the better.

6) Correlation Coefficient: The correlation coefficient (CC) measures the degree of linear correlation between two images and is defined as follows:

$$\text{CC}(X, F) = \sum_{b=1}^{B}\frac{\sum_{i=1}^{\text{HW}}\left(x_i - \mu_{X_i}\right)\left(f_i - \mu_{F_i}\right)}{\sqrt{\sum_{i=1}^{\text{HW}}\left(x_i - \mu_{X_i}\right)^2\sum_{i=1}^{\text{HW}}\left(f_i - \mu_{F_i}\right)^2}} \tag{13}$$

where x_i and f_i represent pixels in images X and F, respectively, and μ_{X_i} and μ_{F_i} are the mean of X_i and F_i, respectively. The closer the CC is to 1, the more relevant the two images are.

7) Universal Image Quality Index: The universal image quality index (UIQI) considers three aspects, namely, correlation loss, luminance aberration, and contrast aberration, and is therefore defined as

$$\text{UIQI}(X, F) = \frac{1}{B}\sum_{i=1}^{B}\frac{\sigma_{X_iF_i}}{\sigma_{X_i}\sigma_{F_i}}\frac{2\mu_{X_i}\mu_{F_i}}{\mu_{X_i}^2 + \mu_{F_i}^2}\frac{2\sigma_{X_i}\sigma_{F_i}}{\sigma_{X_i}^2 + \sigma_{F_i}^2} \tag{14}$$

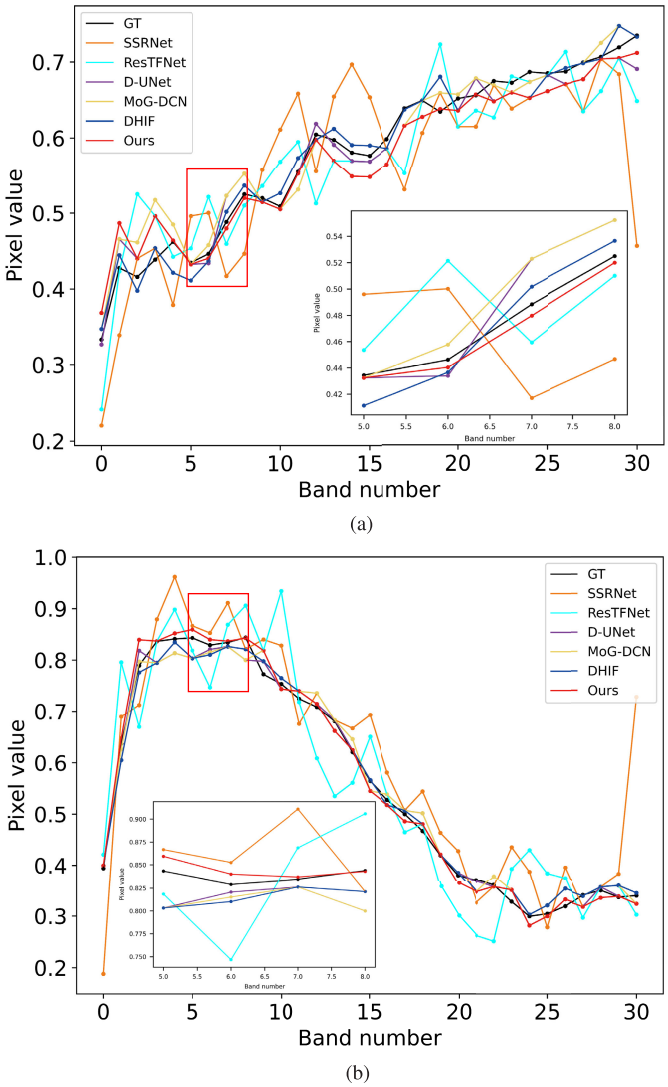


Fig. 8. Spectral vectors of GT and various methods: (a) jelly_beans and (b) stuffed_toys.

where the symbols have the same meaning as in (11). The closer the UIQI is to 1, the better the performance.

D. Results on the CAVE Dataset

In order to intuitively compare the fusion effects of the different methods, we performed a qualitative comparison on the CAVE dataset. The comparison results on two images (jelly_beans and stuffed_toys) are shown in Fig. 7. We show not only the fusion results using the true-color representation but also the residual plots between the fusion results and the ground truth in the form of heat maps. It can be seen that the fusion result of our method is not only the closest to the ground truth in terms of color and texture but also has the smallest residual difference from the ground truth. Especially for the part in the red box, our method only has a small residual difference with the ground truth at the edges, while other methods have a large residual difference not only at the edges but also at the contents. In addition, we show the spectral vector at position (200, 200) in Fig. 8. For a clearer presentation, we zoomed in on the spectral vectors of the four bands (from the fifth to the eighth). It is clear to see that in

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TABLE II
QUANTITATIVE EVALUATION AND PARAMETERS OF DIFFERENT METHODS ON 12 IMAGE PAIRS IN THE CAVE DATASET. m MEANS MILLIONS (RED: OPTIMAL AND BLUE: SUBOPTIMAL)

Methods	PSNR	SAM	ERGAS	SSIM	RMSE	CC	UIQI	#params
FUSE [48]	32.5701	6.4726	6.4078	0.9105	0.0251	0.9815	0.9824	/
GLP-HS [49]	38.6925	6.7664	3.0138	0.9644	0.0125	0.9955	0.9962	/
CSTF [32]	43.0177	6.6897	1.7917	0.9733	0.0077	0.9978	0.9986	/
ResTFNet [50]	44.9044	3.5215	1.5188	0.9908	0.0062	0.9988	0.9990	2.26M
SSRNet [51]	44.7645	3.6530	1.5507	0.9905	0.0095	0.9972	0.9978	0.03M
D-UNet [21]	48.8737	2.4296	0.9894	0.9953	0.0040	0.9994	0.9995	7.23M
MoG-DCN [40]	49.3832	2.2822	0.9436	0.9956	0.0039	0.9994	0.9996	47.28M
DHIF-Net [41]	49.5661	2.2319	0.9202	0.9958	0.0039	0.9994	0.9996	22.46M
MSST-Net [39]	49.7745	2.1567	0.8990	0.9961	0.0037	0.9994	0.9996	32.61M
Ours	49.9503	2.1407	0.8644	0.9962	0.0036	0.9994	0.9996	6.05M

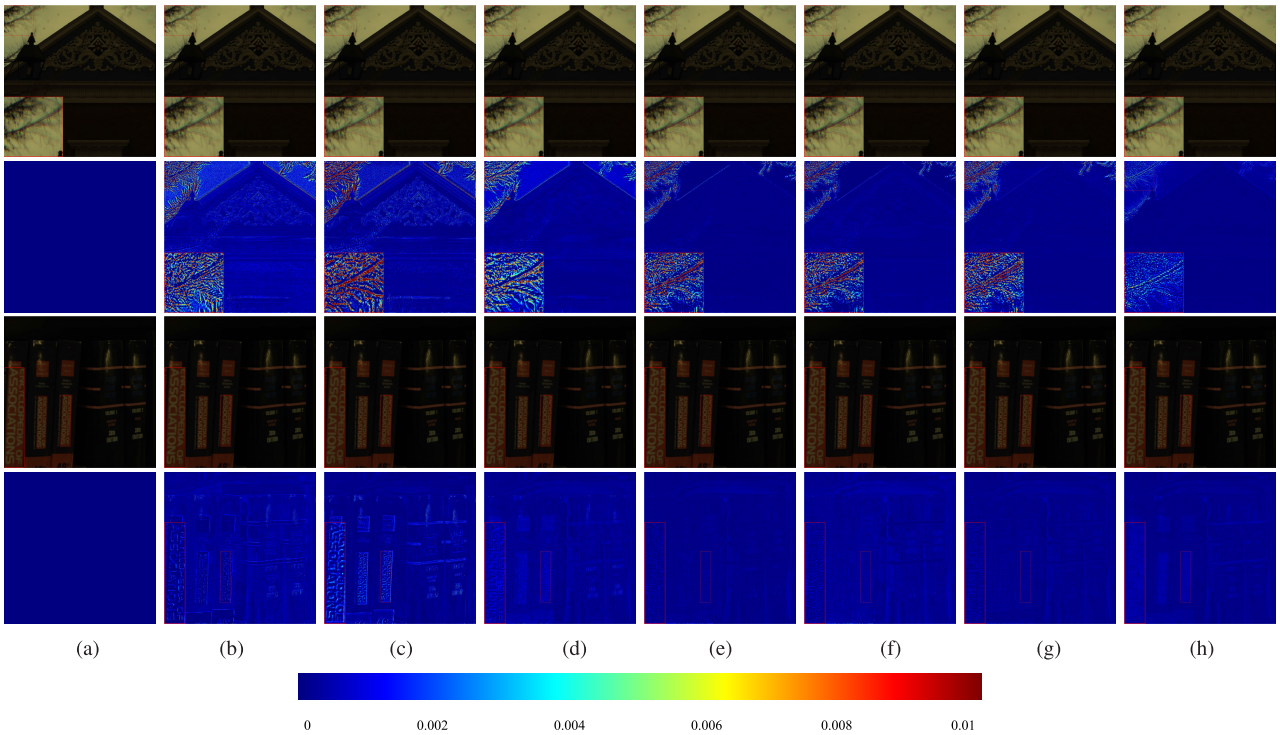


Fig. 9. Qualitative comparison of different methods on the Harvard dataset. The first and third rows show the results using the true-color representation on "imge1" and "imgh2," respectively. The second and fourth rows show the residuals between the GT and the fused results. (a) GT. (b) ResTFNet. (c) SSRNet. (d) D-UNet. (e) MoG-DCN. (f) DHIF-Net. (g) MSST-Net. (h) Ours.

the spectral dimension, the spectral vector of our method is also closest to the ground truth.

As shown in Table II, we show the performance of each fusion algorithm on PSNR, SAM, ERGAS, SSIM, RMSE, CC, and UIQI metrics and the number of parameters for the deep learning-based methods among them. The larger PSNR and the smaller RMSE indicate that our fusion results are closer to the ground truth at the pixel level. In addition, we not only achieve the best results on the SAM and ERGAS metrics, which are specifically used to assess the reconstruction quality of remote sensing images, but also on some generalized metrics in the field of image fusion, such as SSIM, CC, and UIQI. It is particularly noteworthy that our method is able to far outperform other methods in terms of quantitative results while maintaining a smaller number of parameters when compared to methods with a larger number of parameters, such as MSST-Net and MoG-DCN. This is due to the grouped convolution used in the SpePy module. When the number of groups increases, the number of convolution kernels remains the same, but the size of the convolution kernels decreases, so the number of parameters and the computational cost of a convolution decreases by a factor equal to the number of groups.

E. Results on the Harvard Dataset

We also perform a qualitative comparison and a quantitative comparison on the Harvard dataset. Considering that the Harvard dataset has not only indoor scenes but also outdoor scenes, we show in Fig. 9 the fusion results of outdoor scenes (imge1) and indoor scenes (imgh2) separately in order to

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TABLE III
QUANTITATIVE EVALUATION AND PARAMETERS OF DIFFERENT METHODS ON 20 IMAGE PAIRS IN THE HARVARD DATASET. m MEANS MILLIONS (RED: OPTIMAL AND BLUE: SUBOPTIMAL)

Methods	PSNR	SAM	ERGAS	SSIM	RMSE	CC	UIQI	#params
FUSE [48]	35.0849	4.2400	5.9332	0.9054	0.0188	0.9477	0.9688	/
GLP-HS [49]	41.1539	4.3808	3.5904	0.9729	0.0095	0.9799	0.9931	/
CSTF [32]	42.9873	5.2782	3.4275	0.9655	0.0080	0.9797	0.9942	/
ResTFNet [50]	44.4767	3.3502	2.6945	0.9812	0.0069	0.9840	0.9956	2.26M
SSRNet [51]	44.1099	3.3886	2.8350	0.9801	0.0071	0.9827	0.9952	0.03M
D-UNet [21]	47.2373	2.8074	2.1537	0.9889	0.0051	0.9879	0.9973	7.23M
MoG-DCN [40]	48.0329	2.7122	1.9639	0.9898	0.0048	0.9892	0.9977	47.28M
DHIF-Net [41]	48.1848	2.6843	1.8992	0.9901	0.0047	0.9894	0.9977	22.46M
MSST-Net [39]	48.5148	2.6053	1.7989	0.9906	0.0046	0.9901	0.9978	32.61M
Ours	48.6004	2.6005	1.7589	0.9907	0.0045	0.9903	0.9979	6.05M

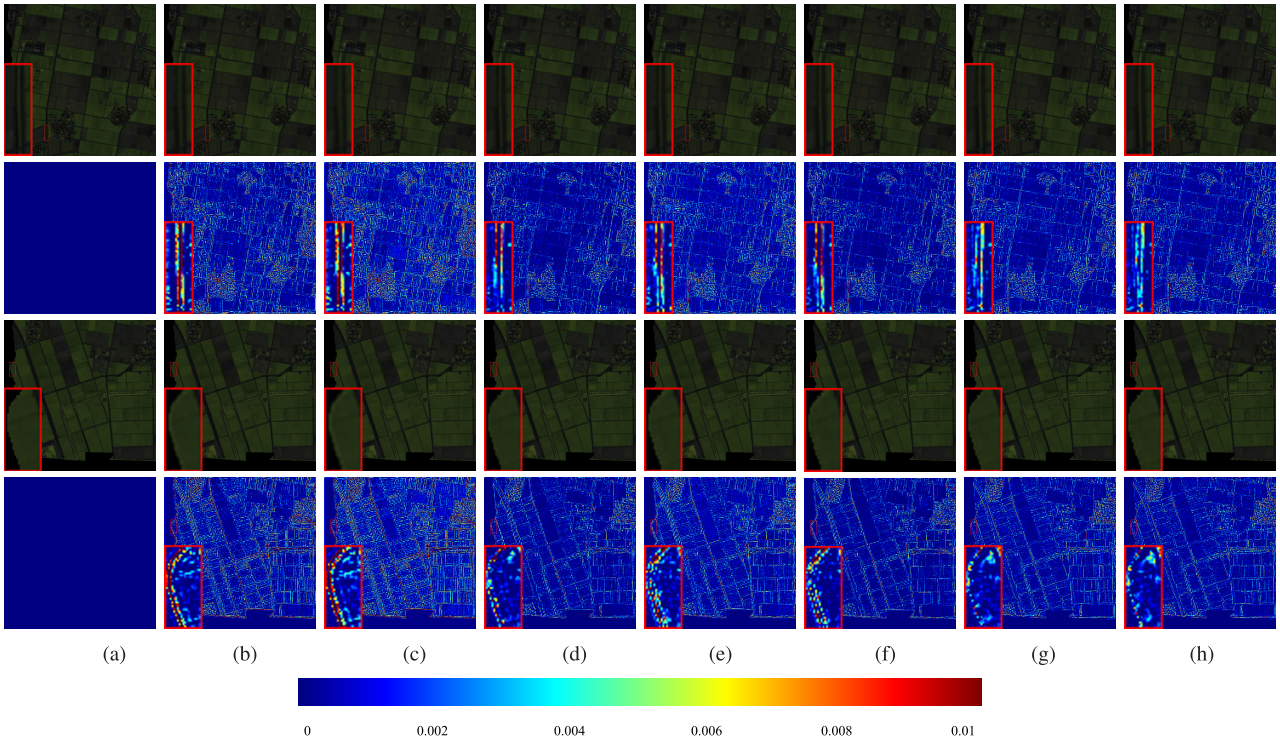


Fig. 10. Qualitative comparison of different methods on the Chikusei dataset. The first and third rows show the results using the pseudocolor representation on "Area 1" and "Area 2," respectively. The second and fourth rows show the residuals between the GT and the fused results. (a) GT. (b) ResTFNet. (c) SSRNet. (d) D-UNet. (e) MoG-DCN. (f) DHIF-Net. (g) MSST-Net. (h) Ours.

fully demonstrate the effect of each algorithm. The outdoor scenes are more challenging compared to the indoor scenes. This is because outdoor scenes are easily disturbed by the outside world and change during the shooting process, such as the tree branches in the red box of imge1. This not only makes spatial distortion between the HSI and MSI but also makes the HSI itself spatially different between the spectra. The existing fusion methods show limitations here and the fusion results are not satisfactory. With the specially designed pyramidal multiscale structure and spectralwise multihead hybrid-attention mechanism, our method is able to face such scenes more robustly by fusing local spectral information to achieve better results. In imgh2, our method is also able to recover the details on books, such as the names on the books, more clearly.

The quantitative comparison results in Table III also demonstrate the superiority of our method. Our method performs best in all indicators. Although our method is only marginally ahead of MSST-Net in all metrics, the number of parameters of our method is much smaller than that of MSST-Net, which is only 18.55% of the number of parameters of MSST-Net. Therefore, our method is more advantageous in practical applications. Compared with DHIF-Net, the third-best method, the indexes of our method are even far superior, with 0.86% improvement in PSNR, 3.12% improvement in SAM, and 7.39% in ERGAS.

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TABLE IV
QUANTITATIVE EVALUATION AND PARAMETERS OF DIFFERENT METHODS ON EIGHT IMAGE PAIRS IN THE CHIKUSEI DATASET. m MEANS MILLIONS (RED: OPTIMAL AND BLUE: SUBOPTIMAL)

Methods	PSNR	SAM	ERGAS	SSIM	RMSE	CC	UIQI	#params
FUSE [48]	36.2537	3.6225	7.4157	0.8976	0.0154	0.8624	0.9718	/
GLP-HS [49]	36.8167	3.1021	5.2840	0.9117	0.0145	0.9272	0.9754	/
CSTF [32]	37.4614	2.7210	3.6439	0.9177	0.0134	0.9574	0.9795	/
ResTFNet [50]	43.7782	1.9736	3.0270	0.9789	0.0066	0.9516	0.9950	2.35M
SSRNet [51]	43.0709	2.2425	3.3326	0.9757	0.0072	0.9463	0.9941	0.44M
D-UNet [21]	46.4524	1.3006	2.0684	0.9884	0.0050	0.9732	0.9972	7.29M
MoG-DCN [40]	46.3251	1.2961	2.0737	0.9880	0.0050	0.9741	0.9971	53.19M
DHIF-Net [41]	46.5397	1.2721	2.0013	0.9885	0.0049	0.9744	0.9972	48.51M
MSST-Net [39]	46.7146	1.1763	1.8202	0.9891	0.0048	0.9782	0.9973	50.72M
Ours	46.7692	1.1723	1.8208	0.9892	0.0048	0.9783	0.9974	6.17M

F. Results on the Chikusei Dataset

To compare the fusion algorithms more comprehensively, we also conducted experiments on the real remote sensing image. We crop eight patches on the Chikusei dataset as the test dataset; the qualitative results for the first two patches are shown in Fig. 10, and the quantitative results for all patches are shown in Table IV. We present the pseudocolor representations of the fused products by combining the 119th, the 90th, and the 69th bands. The remote sensing scenes are so large and complex that existing fusion methods struggle to be applied to them. We cope with such complex scenes by capturing nonlocal information. Looking at the fusion results as a whole, our fusion results have smaller residual values with the ground truth, i.e., closer to the ground truth. Also, the heat maps in the red box also show that our method has better recovery on detailed textures and edges.

Despite the fact that the performance of all methods decreases on real remote sensing datasets due to the significant increase in the number of bands and the decrease in the spatial resolution of the images, our method still demonstrates good performance. The better metrics in Table IV confirm the effectiveness of our approach. Our method achieved optimal results on the six metrics and suboptimal results on the ERGAS metrics. However, our method is only 0.03% lower on the ERGAS metrics. The results on the Chikusei dataset demonstrate the ability of our methods to handle datasets with a large number of spectral bands, in which some targeted designs, such as the SpePy module, work well.

G. Results on the Pavia Centre Dataset

The results of another real remote sensing dataset are shown in Fig. 11 and Table V. We present the pseudocolor representations of the fused products by combining the 64th, the 32nd, and the 11th bands. It is obvious to see that our method has the least difference with the ground truth both in the low-frequency part and the high-frequency part. This is because the injection of nonlocal information makes it easier for the network to focus on the recurring content in the scene and thus complete the high-resolution image reconstruction more accurately. For example, the roofs that appear in abundance in Area 1 and the steps in Area 2 have repetitive feature details. Our method performs much better in these regions and is closest to ground truth.

The results in Table V once again confirm the effectiveness of our method on real remote sensing datasets. The Pavia Centre dataset has a smaller image size compared to the Chikusei dataset, and therefore, the training samples for this dataset are smaller, which causes these image fusion methods to perform worse on this dataset than on the Chikusei dataset. Although both our method and MSST-Net employ the Transformer and apply some multiscale designs in the spectral and spatial dimensions, our method achieves a more concise design by applying the Transformer to the fusion step. It can be seen that our method still achieves the best results while keeping the number of parameters low.

H. Experiments With a Scaling Factor of 8

We performed experiments on the CAVE dataset with a scaling factor of 8 to verify the fusion effect of various methods at different scaling scales. From Table VI, we can see that our method still maintains fairly good competitiveness with the best performance in all seven metrics. In addition, we evaluated the efficiency of these methods. We trained and tested all the methods on the same platform. The training time and testing time are shown in Table VI. MoG-DCN and DHIF-Net have longer training time and testing time because they are unfolding networks, which usually contain four stages. Since the computation of Transformer is time-consuming, Transformer-based methods, such as MSST-Net and ours, are not advantageous in terms of training time and testing time. However, the effect of our method is far better than them, so the efficiency is acceptable.

I. Ablation Study

We validate the effectiveness of our proposed SpePy module, SpaPy module, and MLSIF module by ablation experiments. For brevity and fairness, we conduct the experiments

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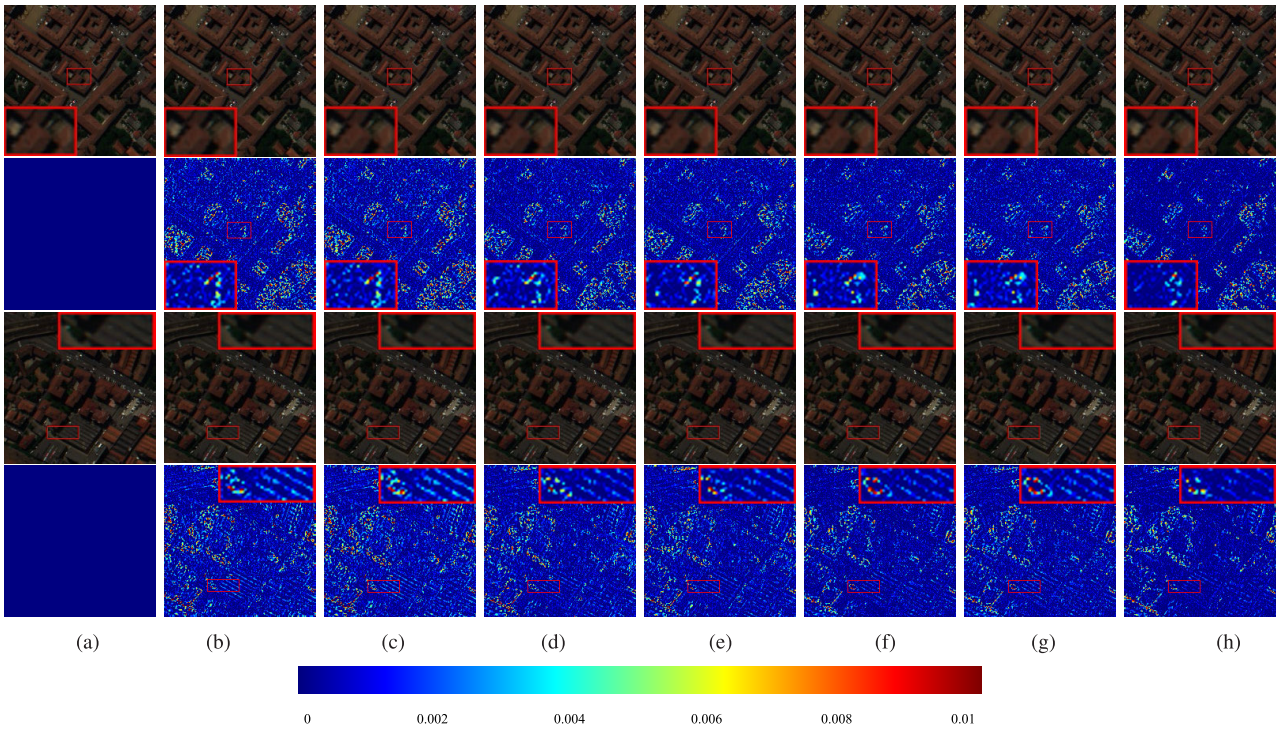


Fig. 11. Qualitative comparison of different methods on the Pavia Centre dataset. The first and third rows show the results using the pseudocolor representation on "Area 1" and "Area 2," respectively. The second and fourth rows show the residuals between the GT and the fused results. (a) GT. (b) ResTFNet. (c) SSRNet. (d) D-UNet. (e) MoG-DCN. (f) DHIF-Net. (g) MSST-Net. (h) Ours.

TABLE V

QUANTITATIVE EVALUATION AND PARAMETERS OF DIFFERENT METHODS ON SIX IMAGE PAIRS IN THE PAVIA CENTRE DATASET. m MEANS MILLIONS (RED: OPTIMAL AND BLUE: SUBOPTIMAL)

Methods	PSNR	SAM	ERGAS	SSIM	RMSE	CC	UIQI	#params
FUSE [48]	26.7286	6.9897	7.9689	0.6691	0.0462	0.8176	0.8639	/
GLP-HS [49]	29.4047	6.4089	4.8217	0.8511	0.0339	0.9500	0.9324	/
CSTF [32]	29.0711	8.1839	4.3242	0.8961	0.0356	0.9722	0.9389	/
ResTFNet [50]	36.8462	3.7066	2.2952	0.9656	0.0145	0.9686	0.9892	2.32M
SSRNet [51]	36.7598	3.7414	2.3145	0.9651	0.0147	0.9685	0.9889	0.28M
D-UNet [21]	37.8862	3.3677	2.1508	0.9709	0.0129	0.9689	0.9914	7.28M
MoG-DCN [40]	38.0727	3.3430	2.1146	0.9717	0.0126	0.9695	0.9918	51.69M
DHIF-Net [41]	38.4963	3.4746	2.1772	0.9705	0.0120	0.9601	0.9926	38.64M
MSST-Net [39]	38.5173	3.0981	1.9708	0.9742	0.0120	0.9717	0.9926	42.36M
Ours	38.5533	3.0896	1.9681	0.9743	0.0119	0.9727	0.9927	6.14M

on the CAVE dataset using four representative metrics, PSNR, SAM, ERGAS, and SSIM, with the scaling factor set to 4.

1) Spectral Branch: We designed a spectral branch consisting of SpePy modules to fully exploit the rich spectral information in HSI. To verify the necessity of the spectral branch, we remove the spectral branch from DSPNet and retrain the network. The results are shown in Table VII. Compared with the fusion results of DSPNet, the fusion results of the network without the spectral branch perform poorly in terms of objective evaluation metrics.

2) Spatial Branch: In the same way, we verify the significance by ablating the spatial branch in the network. Without the spatial branch, the network can only extract spatial information from the spatial–spectral branch. The quantitative results are shown in Table VII. The lower evaluation metrics suggest that this is not enough to recover high-frequency spatial information of HSIs.

3) MLSIF Module: We add the MLSIF module for fusing multiscale local spectral information to the spatial–spectral branch. We hope to use the attention mechanism in it to establish long-range dependencies in the spectral dimension for better fusion of spectral information. To validate our idea, we choose to replace the MLSIF module with the operation of summing multiscale local spectral information. The quantitative results in Table VII show that the introduction of the attention mechanism is beneficial for the fusion of spectral information.

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TABLE VI

QUANTITATIVE EVALUATION, TRAINING TIME, AND TESTING TIME OF DIFFERENT METHODS ON 12 IMAGE PAIRS IN THE CAVE DATASET ON A SCALING FACTOR OF 8. h MEANS HOURS. s MEANS SECONDS (RED: OPTIMAL AND BLUE: SUBOPTIMAL)

Methods	PSNR	SAM	ERGAS	SSIM	RMSE	CC	UIQI	training time	testing time
FUSE [48]	30.6403	7.5171	3.9902	0.8769	0.0311	0.9711	0.9731	-	43.86s
GLP-HS [49]	40.2715	7.2969	1.2516	0.9691	0.0104	0.9968	0.9974	-	81.35s
CSTF [32]	42.5134	6.5432	0.9656	0.9766	0.0079	0.9974	0.9984	-	301.51s
ResTFNet [50]	43.8966	3.7200	0.8551	0.9883	0.0070	0.9982	0.9987	3h	3.77s
SSRNet [51]	43.5659	3.8372	0.8879	0.9871	0.0072	0.9981	0.9985	4h	3.79s
D-UNet [21]	48.2904	2.4041	0.5223	0.9952	0.0044	0.9992	0.9994	10.5h	3.83s
MoG-DCN [40]	48.6618	2.2670	0.5081	0.9954	0.0043	0.9992	0.9994	20h	7.91s
DHIF-Net [41]	48.7259	2.2544	0.5062	0.9955	0.0043	0.9992	0.9994	22.5h	8.10s
MSST-Net [39]	48.9116	2.2368	0.4967	0.9956	0.0042	0.9993	0.9994	30h	5.16s
Ours	49.0819	2.2093	0.4891	0.9957	0.0041	0.9993	0.9995	25h	4.07s

TABLE VII

QUANTITATIVE EVALUATION RESULTS OF ABLATION ON THE CAVE DATASET. THE W/O SPE-BRANCH, W/O SPA-BRANCH, AND W/O MLSIF INDICATE DSPNET WITHOUT THE SPECTRAL BRANCH, DSPNET WITHOUT THE SPATIAL BRANCH, AND DSPNET WITHOUT THE MLSIF MODULE, RESPECTIVELY (**BOLD**: OPTIMAL)

Methods	PSNR	SAM	ERGAS	SSIM
w/o spe-branch	48.9499	2.3389	0.9809	0.9955
w/o spa-branch	48.5287	2.4138	1.0240	0.9952
w/o MLSIF	49.2453	2.2837	0.9476	0.9957
Ours	**49.9503**	**2.1407**	**0.8644**	**0.9962**

TABLE VIII

QUANTITATIVE EVALUATION RESULTS OF ABLATION ON THE CAVE DATASET. THE SPAPY + CONV AND SPAPY + Gi ($i \in \{2, 4, 8\}$) ARE THE RESULTS OF THE ABLATION EXPERIMENTS ON THE SPEPY MODULE. THE SPEPY + CONV IS THE RESULT OF THE ABLATION EXPERIMENT ON THE SPAPY MODULE (**BOLD**: OPTIMAL)

Methods	PSNR	SAM	ERGAS	SSIM
SpaPy + conv	49.4850	2.2392	0.9188	0.9958
SpaPy + G2	49.5401	2.2335	0.9116	0.9959
SpaPy + G4	49.6665	2.2165	0.8984	0.9959
SpaPy + G8	49.8289	2.1674	0.8783	0.9960
SpePy + conv	49.6494	2.2974	0.9035	0.9959
Ours	**49.9503**	**2.1407**	**0.8644**	**0.9962**

4) SpePy Module: The SpePy module is able to extract multiscale local spectral information, which is different from other methods to extract global spatial–spectral information of HSI directly. We verify the importance of multiscale local spectral information in two ways: first, by removing the band grouping operation to verify the importance of local information, and second, by keeping only one type of grouping to verify the importance of multiscale. In addition, we keep the parameter number of subsequent convolutional layers constant to ensure the fairness of the experiment. The results of the ablation experiments are shown in Table VIII, where SpaPy + conv shows that we remove the band grouping operation from the SpePy module, and SpaPy + G2, SpaPy + G4, and SpaPy + G8 indicate that the bands are divided into only two, four, and eight groups, respectively. It is clear that removing the band grouping operation leads to a decrease in performance, which indicates that the band grouping operation facilitates a finer reconstruction of the HSI. In addition, although the grouping method "dividing the bands into eight groups only" performs better than the other two grouping methods, it is still inferior to our proposed method, suggesting that multiscale information is better than any of the single-scale information. This is because although the adjacent bands are more correlated, viewing the spectra from a larger scale facilitates the network to better capture the trends of spectral variations. The combination of multiscale information allows the network to more fully reconstruct the spectral information.

5) SpaPy Module: Similarly, the design for extracting multiscale spatial information also exists in the SpaPy module. We extract the nonlocal spatial information with the help of a downsampling operation. To verify the effect, we replace the whole SpaPy module with a double convolutional layer and show the results in Table VIII, denoted by SpePy + conv. The experimental results show that the multiscale design is necessary for the extraction of spatial information.

V. CONCLUSION

In this article, we propose a three-branch MSI and HSI fusion network, named DSPNet. Multiscale design is applied to both spectral dimension and spatial dimension. To fully exploit the spectral information of the HSI, the SpePy module of the spectral branch extracts the multiscale local spectral information with the help of band grouping operation. The MLSIF module of the spatial–spectral branch integrates local spectral information more efficiently by establishing long-range dependencies between local spectral information in the spectral dimension. Similarly, the SpaPy module of the spatial branch better copes with complex remote sensing scenarios by extracting nonlocal spatial information. The ablation experiments verify the effectiveness of the above design.

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Compared with other state-of-the-art fusion methods, our method can achieve advanced performance both qualitatively and quantitatively.

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